

EXPERIMENT 10 – Paging Technique

Aim: To implement the Paging Technique and perform logical-to-physical address translation.

C Program:

```
#include <stdio.h>

int main()
{
    int pageTable[20];
    int pageSize, numPages, logicalAddress;
    int pageNumber, offset, frameNumber, physicalAddress, i;

    printf("Enter Page Size: ");
    scanf("%d", &pageSize);

    printf("Enter Number of Pages: ");
    scanf("%d", &numPages);

    printf("Enter Frame Numbers for Each Page:\n");
    for(i = 0; i < numPages; i++)
    {
        printf("Page %d -> Frame: ", i);
        scanf("%d", &pageTable[i]);
    }

    printf("Enter Logical Address: ");
    scanf("%d", &logicalAddress);

    pageNumber = logicalAddress / pageSize;
    offset = logicalAddress % pageSize;

    if(pageNumber >= numPages)
    {
        printf("Invalid Logical Address\n");
        return 0;
    }

    frameNumber = pageTable[pageNumber];
    physicalAddress = (frameNumber * pageSize) + offset;

    printf("\nPage Number    : %d", pageNumber);
    printf("\nOffset          : %d", offset);
    printf("\nFrame Number    : %d", frameNumber);
    printf("\nPhysical Address : %d\n", physicalAddress);

    return 0;
}
```

Compilation: `gcc paging.c -o paging``

Output (Page Size=256, 4 Pages, Frame Table=[5,10,3,7], Logical Address=660):

```
Page Number    : 2
Offset         : 148
Frame Number    : 3
Physical Address : 916
```

Shell Script:

```
#!/bin/bash

echo "Enter Page Size:"
read pageSize

echo "Enter Number of Pages:"
read numPages

declare -a pageTable

for ((i=0;i<numPages;i++))
do
    echo "Enter Frame Number for Page $i:"
    read pageTable[$i]
done

echo "Enter Logical Address:"
read logicalAddress

pageNumber=$((logicalAddress / pageSize))
```

```
offset=$((logicalAddress % pageSize))

if [ $pageNumber -ge $numPages ]
then
    echo "Invalid Logical Address"
    exit
fi

frameNumber=${pageTable[$pageNumber]}
physicalAddress=$((frameNumber * pageSize + offset))

echo "Page Number      : $pageNumber"
echo "Offset           : $offset"
echo "Frame Number      : $frameNumber"
echo "Physical Address : $physicalAddress"
```

Result: The Paging Technique was implemented and address translation verified.
